6 Months Training TR-104 Full Stack Development

January, 2026

Day 13 – Route Guards and Role-Based Access Control

1. Objective of the Day

The objective of the day was to implement authentication and authorization using Angular route guards and enforce role-based access control within the application.

2. Tasks Performed

- Implemented Authentication Guard (AuthGuard) to protect routes
- Restricted access to protected routes for unauthenticated users
- Redirected unauthorized users to the login page
- Implemented Authorization Guard (AdminGuard) for role-based access
- Restricted admin routes to users with admin role only
- Managed user roles using authentication service
- Applied guards using canActivate in routing configuration
- Controlled UI elements based on authentication state
- Displayed login, logout, and admin navigation options dynamically
- Implemented logout functionality and state reset

3. Tools / Technologies Used

- Angular Framework
- TypeScript
- Angular Routing

4. Key Learnings

- Route guards control access before component loading
- Authentication and authorization are different concepts
- Role-based access improves application security
- Services help manage user roles and authentication state
- UI should reflect authentication and authorization state

5. Challenges Faced

- Understanding guard implementation and routing configuration
- Managing role-based logic correctly

6. Solutions

- Practiced guard implementation step-by-step
- Tested different user roles for proper access control

7. Conclusion

- Successfully implemented route protection and role-based access
- Strengthened application security and navigation control